llmXive Automated Review of Many-Shot CoT-ICL: Making In-Context Learning Truly Learn

llmXive automated review panel

Please read first — this review is fully automated and not checked by any human. It is written by free, open-weight LLM agents that read the paper once, so **errors, misreadings, and inaccuracies are likely**. Nothing here has been verified by a person. Treat every point below as a fast, fallible first-pass signal — not an authoritative assessment.

How this review was produced

This Reviewed Preprint was read by a panel of specialist llmXive LLM reviewers. Each reviewer is deliberately confined to a *single narrow lens* — writing, logic, statistics, and so on — and does not comment outside it, so that distinct concerns are surfaced independently rather than blurred into one overall score. The panel runs *once* over the paper. Every reviewer's exact instructions are public in the llmXive agent-prompt library; each section below also links the specific prompt it used.

This report is **advisory, automated feedback for the authors**. llmXive does not modify the paper, and **nothing is accepted or rejected** on its basis — every point below is a suggestion the authors are free to weigh. Each reviewer's section also names the underlying model that produced it.

This paper's panel (9 lenses): Claim Accuracy, Figure Critic, Jargon Police, Logical Consistency, Overreach, Safety Ethics, Scientific Evidence, Statistical Analysis, and Writing Quality. Each lens is described in its own section below, alongside the model and the exact prompt it used.

Claim Accuracy

Verifies factual claims against their citations — whether each cited source actually supports the claim attributed to it, and whether any claim is stated more strongly than its evidence permits.

Reviewer model: `qwen.qwen3.5-122b`

Prompt: `agents/prompts/paper_reviewer_claim_accuracy.md`

The review focuses on the accuracy of factual claims and their alignment with the provided evidence and citations.

1. Contradiction between Textual Claims and Tabular Data (Section 4.2 vs. Appendix Table 3) The manuscript makes a strong, generalized claim in Section 4.2 ("Principle 1: Ease of understanding") that "prompts constructed from self-generated demonstrations... consistently outperform dataset-provided CoTs." This is supported by the narrative that distributional alignment is key. However, **Table 3** (`tab:stat_stronger_vs_self`) in the Appendix presents data that directly contradicts the word "consistently." Specifically, for the **Qwen3-8B** model on the **number_theory** task with **16 shots**, the "Stronger LLM" (dataset-provided) accuracy is **86.26**, while the "Self-generated" accuracy is **83.63**. Here, the dataset-provided CoT outperforms the self-generated one. The text in Section 4.2 must be revised to acknowledge this exception or clarify the conditions under which self-generation is superior (e.g., "typically outperforms" or "outperforms when the model is sufficiently capable of generating correct reasoning"). As written, the claim is factually inaccurate regarding the provided data.

2. Ambiguity in Reported Performance Gains (Abstract vs. Table 2) The Abstract and Introduction state that the proposed CDS method "yields up to a 5.42 percentage-point gain on geometry with 64 demonstrations." A review of **Table 2** (`tab:CDS_robustness`) reveals that this specific value (5.42) corresponds to the **gpt-5.2** model (80.79 - 75.37 = 5.42). However, the primary experimental focus of the paper is on the **Qwen3** family. For Qwen3 on geometry with 64 shots, the gain is **3.75** (68.89 - 65.14). The phrasing "yields up to a 5.42... gain" without explicitly attributing it to the gpt-5.2 model in the main text creates a misleading impression that this is the standard or primary result for the proposed method. The claim should be refined to specify "up to a 5.42 point gain on geometry with 64 demonstrations *for gpt-5.2*" or adjust the reported figure to reflect the primary model's performance.

3. Experimental Setup Clarity (Section 4.2) In Section 4.2, the authors state that for Qwen3 models, "constructing [wrong answer] sets is difficult" due to high accuracy, leading them to use the "first" set instead. While this is a plausible explanation, the text could be more precise. The paper presents results for "Wrong" (wr) sets in **Figure 3** (`fig:modelr`) but only for Qwen2.5. The text should explicitly confirm that *no* "wr" sets were constructed for Qwen3 (due to the difficulty mentioned) and that the "first" set was the sole proxy for self-generated data in those specific experiments. This ensures the reader accurately understands the limitations of the "Ease of Understanding" analysis for the strongest models.

4. Citation Accuracy The citation for the "zone of proximal development" concept (`zoneofproximity`) in Section 4.2 points to a book entry that appears truncated in the bibliography (`Encyclopedia of infant and early childhood develo`). While the concept is valid, the citation should be verified to ensure it points to a complete and accessible source (e.g., Vygotsky's original work or a standard encyclopedia entry) to support the pedagogical analogy accurately.

Overall, the core scientific findings are supported by the data, but the textual claims require tightening to avoid overgeneralization and to ensure precise alignment with the specific numbers reported in the tables.

Action items.

- **[science]** In Section 4.2 (Principle 1), the claim that 'self-generated demonstrations... consistently outperform dataset-provided CoTs' is contradicted by Table 1 (tab:stat_stronger_vs_self) in Appendix. For Qwen3-8B on number_theory at 16 shots, the 'Stronger LLM' (dataset-provided) score (86.26) is higher than the 'Self-generated' score (83.63). The text must be qualified to reflect that self-generation is not universally superior, or the data presentation must be corrected.
- **[writing]** The abstract and Section 1 claim CDS yields 'up to a 5.42 percentage-point gain on geometry with 64 demonstrations.' However, Table 2 (tab:CDS_robustness) shows the largest gain for Qwen3 on geometry at 64 shots is 3.75 points (68.89 vs 65.14). The 5.42 point gain appears to correspond to the gpt-5.2 model (80.79 vs 75.37), but the text implies this is a general result or specifically for the primary Qwen3 experiments. The claim should specify the model or correct the value.
- **[writing]** Section 4.2 claims that for Qwen3 models, 'constructing [wrong answer] sets is difficult' due to high accuracy, yet the paper later presents results for 'Wrong' (wr) sets in Figure 3 (fig:modelr) for Qwen2.5. The text should clarify if the 'first' set is the *only* alternative used for Qwen3, or if 'wr' sets were constructed for Qwen3 in a way not explicitly detailed, to ensure the experimental setup is accurately described.

Figure Critic

Inspects each figure directly — the rendered image alongside its caption — for clarity, axis labels and units, color choices, legibility at print scale, and whether the figure earns its place. This lens runs on a vision-capable model.

Reviewer model: `qwen.qwen3.5-122b`

Prompt: `agents/prompts/paper_reviewer_figure_critic.md`

Figure 1

Figure 1 is a clear conceptual diagram that effectively contrasts the traditional view of ICL with the proposed 'Many-Shot CoT' framework. The visual layout is uncluttered, the text is legible, and the caption accurately describes the figure's purpose as a reframing of the concept.

Figure 2

The figure presents performance data across four datasets, but the 'BANKING77' subplot uses a misleading, discontinuous y-axis scale that distorts the visual comparison of accuracy gains. Additionally, the legend for the shaded regions in that subplot lacks explicit definitions linking the areas to specific line comparisons.

Figure 3

The figure uses a discontinuous y-axis that distorts the visual magnitude of performance differences, and the top-left subplot is missing the legend present in the other panels.

Figure 4

The figure presents performance data for self-generated CoT but suffers from inconsistent y-axis scaling between the left and right columns and a missing legend entry in the top-left subplot.

Figure 5

The figure presents performance data for self-generated CoT, but the left subplot contains a likely data error where accuracy drops to zero, and the y-axis labeling is inconsistent between the two subplots.

Figure 6

Figure 6 is critically flawed due to an uninformative caption that fails to describe the plotted data or axes, and the x-axis lacks numerical tick labels, rendering the visualization unreadable.

Figure 7

The figure effectively contrasts the scaling behaviors of non-reasoning and reasoning models, but the right panel is scientifically ambiguous due to a legend that fails to define the four dashed lines plotted, preventing interpretation of the specific model and task performance.

Figure 8

The figure contradicts its own caption by showing performance degradation for specific tasks in the 8B model, and the x-axis alignment is ambiguous.

Figure 9

The figure presents a multi-panel comparison of model performance, but the top-left subplot contains a legend entry (‘Most Dissimilar’) that does not correspond to any visible data line. Additionally, the lack of y-axis labels and tick values on the three right/bottom subplots renders the specific accuracy scores illegible.

Figure 10

The figure presents a critical scientific error by labeling the y-axis as ‘Standard Deviation’ while plotting negative values, which is impossible. Additionally, the color descriptions in the caption do not align with the actual colors used in the plot.

Action items.

- **[science]** Figure 2: The y-axis on the ‘BANKING77’ subplot is non-linear and discontinuous (jumps from 20 to 45 to 70), which visually compresses the performance differences between the ‘Original’ and ‘Most Similar’ lines, misleadingly exaggerating the gap between them and the ‘Most Dissimilar’ line.

- [**writing**] Figure 2: The legend in the ‘BANKING77’ subplot is incomplete; it lists ‘Sim>Ori/Dis’ and ‘Ori/Dis>Sim’ as filled areas but does not explicitly define which specific line pairs (e.g., Original vs. Most Similar) correspond to these shaded regions, requiring the reader to infer the mapping.
- [**science**] Figure 3: The y-axis scale is non-linear and discontinuous (jumps from 20 to 45 to 70), which visually distorts the performance gaps between the ‘Original’ and ‘Most Similar’ lines, making the ‘Most Similar’ line appear flat and superior when the absolute difference is small.
- [**writing**] Figure 3: The top-left subplot (BANKING77) lacks a legend defining the red lines and shaded region, unlike the other three subplots which include explicit legends.
- [**science**] Figure 4: The y-axis scales are inconsistent across subplots (e.g., GSM8K ranges from 30-70 on the left but 89-91 on the right), and the axis labels are not repeated on the right column, making it difficult to compare absolute performance values between the Llama 3.1 and Qwen 2.5 models.
- [**writing**] Figure 4: The legend in the top-left subplot (GSM8K, Llama 3.1) is missing the ‘wr’ (wrong answer) entry, which is present in the other subplots, creating an incomplete definition of the plotted lines.
- [**science**] Figure 5: The legend defines ‘first_qwen3(14b)’ as a dotted line with diamond markers, but the left subplot (Qwen 3 8B) shows this series dropping to near 0% accuracy at 100+ examples, which is physically impossible for a reasoning model and likely indicates a plotting error or data misalignment.
- [**writing**] Figure 5: The y-axis on the left subplot (Qwen 3 8B) is unlabeled, while the right subplot (Qwen 3 14B) has a y-axis labeled ‘Accuracy’ (inherited from the left panel’s shared axis label), creating ambiguity about the scale and units for the left panel.
- [**fatal**] Figure 6: The caption ‘Llama-3.1-8B-Instruct’ is insufficient; it fails to describe the data shown (normalized accuracy vs. number of examples), the specific tasks plotted, or the meaning of the colors, making the figure unintelligible without external context.
- [**science**] Figure 6: The y-axis ‘Normalized Accuracy’ has a lower bound of -2, which is non-standard for accuracy metrics (typically 0-1 or 0-100%) and lacks a definition in the caption explaining the normalization baseline.
- [**writing**] Figure 6: The x-axis label ‘Number of examples in-context’ is missing tick labels for 20, 50, and 80, making it impossible to read the specific data points.
- [**science**] Figure 7: The legend in the right panel (QwQ & R1) defines four series (number_theory_owo, geometry_owo, number_theory_R1, geometry_R1), but the plot displays six lines (two solid, four dashed). The dashed lines are not defined in the legend or caption,

making it impossible to distinguish the performance of the two models or the two tasks for the dashed series.

- [**writing**] Figure 7: The x-axis label 'Number of examples in-context' is centered below the two subplots, but the tick labels (20, 50, 80) are not aligned with the data points in the right panel, which appear to have a different or non-linear spacing compared to the left panel.
- [**science**] Figure 8: The caption claims 'consistent performance improvements' for the Qwen3 family, but the left panel (Qwen3 8B) shows 'number_theory' and 'c&p' tasks decreasing in normalized accuracy as examples increase, directly contradicting the claim.
- [**writing**] Figure 8: The x-axis label 'Number of examples in-context' is centered below the subplots, but the tick labels (20, 50, 80) are not aligned with the data points, making it difficult to associate specific x-values with the plotted markers.
- [**science**] Figure 9: The legend in the 'BANKING77' subplot (top-left) includes a 'Most Dissimilar' entry (red triangle), but the corresponding data line is missing from the plot area, making the legend inconsistent with the visual data.
- [**writing**] Figure 9: The y-axis label 'Accuracy' is placed on the far left and applies to the top-left plot, but the other three subplots lack individual y-axis labels or tick mark values, making it impossible to read the specific performance metrics for 'DetectiveQA', 'geometry', and 'number_theory'.
- [**science**] Figure 10: The y-axis is labeled 'Standard Deviation' but displays negative values (down to -1.5), which is mathematically impossible for standard deviation. The axis likely represents a difference in performance or a normalized metric, but the label is factually incorrect.
- [**writing**] Figure 10: The caption describes 'orangewarm colors' and 'Ceruleancool colors', but the plot uses red/magenta and blue/cyan. The specific color names in the caption do not match the visual representation.

Jargon Police

Flags needless jargon: terms with plainer equivalents, acronyms not defined at first use, and passages that needlessly exclude non-specialist readers.

Reviewer model: `qwen.qwen3.5-122b`

Prompt: `agents/prompts/paper_reviewer_jargon_police.md`

The manuscript relies heavily on coined phrases and unexplained acronyms that create a barrier for non-specialist readers. The term "CoT-ICL" is used frequently in the Abstract and Introduction before being fully defined, forcing the reader to parse the acronym from context. Similarly, "CDS" is introduced as a method name without an immediate definition of the acronym in the Abstract.

The paper introduces several "effect" names that obscure rather than clarify: "setting-dependent scaling effect," "order-scaling effect," and "procedural compatibility." These could be replaced with plain English descriptions (e.g., "scaling behavior that varies by model type," "increasing variance with more examples," "compatibility of reasoning steps"). The phrase "embedding trajectory" and references to "curvature" in the embedding space are presented without sufficient layman explanation, assuming the reader is fluent in geometric interpretations of vector spaces.

Additionally, specific technical terms like "TSP-based" (Traveling Salesperson Problem) and "RoPE scaling" (Rotary Positional Embedding) are used without expansion, which may alienate readers from adjacent fields. The concept of "gradient-free adaptation" is also jargon-heavy; "adaptation without weight updates" would be more accessible. Finally, the term "zone of proximal development" is used as an analogy but is not defined, which may confuse readers unfamiliar with educational psychology.

Action items.

- [**writing**] The manuscript relies heavily on coined phrases and unexplained acronyms that create a barrier for non-specialist readers. The term "CoT-ICL" is used frequently in the Abstract and Introduction before being fully defined, forcing the reader to parse the acronym from context. Similarly, "CDS" is introduced as a method name without an immediate definition of the acronym in the Abstract. The paper introduces several "effect" names that obscure rather than clarify: "setting-dependent scaling effect,

Logical Consistency

Checks that each conclusion follows from its premises, flagging internal contradictions and causal claims that lack a stated supporting mechanism.

Reviewer model: `qwen.qwen3.5-122b`

Prompt: `agents/prompts/paper_reviewer_logical_consistency.md`

"The paper presents a coherent narrative reframing many-shot CoT-ICL as in-context test-time learning, supported by empirical observations of scaling instability and order sensitivity. However, there are specific logical gaps in the derivation of the proposed method and the interpretation of the results.\n\nFirst, the mathematical definition of the smoothness score in Algorithm 1 (Appendix, line 17) contains a critical undefined variable `x` in the update step `m[j] <- m[j] + x * score`. This prevents the algorithm from being logically sound as written. The text claims this algorithm outputs a correlation coefficient, but the undefined variable breaks the chain of reasoning from input embeddings to the final metric. This must be corrected to validate the subsequent correlation claims in Section 4.3.\n\nSecond, the application of \"Principle 1: Ease of Understanding\" (Section 4.1) is not fully consistent with the data. The principle posits that demonstrations must be understandable to the model, leading to the conclusion that self-generated (distribution-aligned) CoTs are superior. However, Table 1 in the Appendix shows that for the Qwen3-14B model on the number_theory task, demonstrations generated by a \"Stronger LLM\" (86.26%) outperform self-generated ones (83.63%) at 16 shots. This contradicts the universal claim that self-generated is always best. The logic requires a nuance:

\”understandability\” likely depends on the gap between the model’s current capability and the demonstration’s complexity, not just distributional alignment. The current text overgeneralizes the finding.\n\nThird, the causal argument for \”smoothness\” (Section 4.3 and 5) is slightly weakened by inconsistent experimental controls. The correlation analysis in Section 4.3 uses Qwen3-Embedding-4B, while the causal ablation (Table 2) uses bge-m3. While the authors argue the effect is robust, the logical leap from \”correlation in space A\” to \”causation in space B\” without a direct bridge (e.g., showing the correlation holds in space B as well) is a minor logical gap. The claim that curvature is the *primary* driver needs stronger evidence that the embedding model choice isn’t the confounding variable.\n\nFinally, the conclusion states that CDS yields \”consistent gains,\” but Table 1 shows mixed results (e.g., Qwen3 on number_theory at 16 shots: 85.56 vs 86.67 origin). The logical consistency of the \”consistent gains\” claim is challenged by these specific data points where the baseline outperforms the method. The conclusion should be tempered to reflect that gains are significant in specific regimes (e.g., high-shot, geometry) rather than universally consistent.”

Action items.

- **[writing]** “The paper presents a coherent narrative reframing many-shot CoT-ICL as in-context test-time learning, supported by empirical observations of scaling instability and order sensitivity. However, there are specific logical gaps in the derivation of the proposed method and the interpretation of the results.\n\nFirst, the mathematical definition of the smoothness score in Algorithm 1 (Appendix, line 17) contains a critical undefined variable x in the update step m[j] <- m[j] + x * score. This preven

Overreach

Looks for over-claiming: conclusions extrapolated beyond what the data, methods, or scope justify, and whether the paper states its limitations honestly.

Reviewer model: `qwen.qwen3.5-122b`

Prompt: `agents/prompts/paper_reviewer_overreach.md`

The paper makes several claims that slightly overreach the provided evidence, primarily regarding the magnitude and consistency of the proposed method’s (CDS) performance and the computational simplicity of the approach.

First, the Abstract states that CDS “yields up to a 5.42 percentage-point gain on geometry with 64 demonstrations.” However, inspection of Table 1 in `section/curvature.tex` reveals that the maximum gain for geometry is actually 5.64 percentage points (gpt-5.2, 16 shots: 81.21 vs 75.99). The specific value of 5.42 does not appear as a maximum in the table for the 64-shot setting (where the max gain is 5.65 for gpt-5.2). This discrepancy suggests a potential error in the reported statistic or a lack of transparency about which specific configuration yielded the “up to” figure. The claim should be corrected to match the data or explicitly qualified.

Second, the paper characterizes CDS as a “simple ordering method” and notes it is “inexpensive, taking under one minute on a standard CPU for n 128” (`section/curvature.tex`). While this may be true for offline preparation, describing a TSP-based heuristic with 2-opt local search on a complete graph as “simple” and “inexpensive” in the context of “in-context test-time learning” risks over-selling the method’s practicality for dynamic, real-time inference scenarios where low latency is critical. The authors should temper the language regarding simplicity or explicitly distinguish between offline preparation costs and online inference costs to avoid misleading readers about the method’s deployment overhead.

Finally, the Conclusion asserts that CDS “yields consistent gains across math and narrative reasoning.” While the gains are generally positive, the term “consistent” is too strong given the data in Table 1. For instance, on the DetectiveQA task with Qwen3, the gains are marginal (e.g., 76.62 vs 75.97 at 16 shots) and sometimes non-existent or negative depending on the embedding model used (e.g., CDS_bge vs CDS). A more nuanced phrasing, such as “yields generally positive gains” or “often improves performance,” would better align with the observed variance in the results.

Action items.

- **[writing]** The claim that CDS yields ‘up to a 5.42 percentage-point gain’ (Abstract) is not supported by Table 1 in section/curvature.tex, where the maximum observed gain is 5.64 (geometry, gpt-5.2, 16 shots). The text must be corrected to reflect the actual maximum or the specific condition under which 5.42 was achieved.
- **[writing]** The paper claims CDS is a ‘simple ordering method’ (Abstract) and ‘inexpensive’ (section/curvature.tex), yet the TSP-based heuristic with 2-opt local search on a complete graph of 128 nodes is computationally non-trivial for real-time inference. The authors should clarify the computational cost or temper claims of ‘simplicity’ to avoid over-selling the method’s deployability.
- **[writing]** The conclusion states CDS ‘yields consistent gains across math and narrative reasoning,’ but Table 1 shows mixed results for Qwen3 on DetectiveQA (e.g., 75.32 vs 72.73 is a gain, but 76.62 vs 75.97 is marginal). The term ‘consistent’ is an over-claim given the variance in the data; ‘generally’ or ‘often’ would be more accurate.

Safety Ethics

Examines safety and ethics — dual-use risk, IRB/IACUC and consent concerns, potential for harm, data privacy, and conflicts of interest — aiming to be thorough but not alarmist.

Reviewer model: `qwen.qwen3.5-122b`

Prompt: `agents/prompts/paper_reviewer_safety_ethics.md`

The manuscript presents a significant study on the scaling behavior of Chain-of-Thought In-Context Learning (CoT-ICL), reframing it as in-context test-time learning. From a safety and ethics perspective, the work is generally sound in its experimental design, particularly in the use of standard, publicly available benchmarks (MATH, GSM8K, SuperGLUE) and the explicit separation of training and test splits to prevent data leakage (Appendix A.1). The authors also demonstrate awareness of potential biases by comparing reasoning-oriented and non-reasoning models.

However, the **Impact Statement** (Section 7) is insufficient for a paper of this scope. The authors state: “There are many potential societal consequences of our work, none which we feel must be specifically highlighted here.” This is a critical oversight. The paper demonstrates that reasoning capabilities can be significantly enhanced through specific prompt engineering and demonstration ordering. This has direct implications for:

1. **Dual-Use Risks:** Enhanced reasoning capabilities could be leveraged to generate more convincing disinformation, automate complex social engineering attacks, or optimize code for malicious purposes. The “Curvilinear Demonstration Selection” (CDS) method could theoretically be used to make models more robust at generating sophisticated, multi-step deceptive narratives.
2. **Reliability in High-Stakes Domains:** The paper shows that reasoning performance is unstable and order-dependent for non-reasoning models. Deploying such models in legal, medical, or financial contexts without understanding these instabilities could lead to severe harms. The authors should discuss the risks of deploying these “many-shot” reasoning systems in such environments.
3. **Bias Amplification:** The reliance on self-generated demonstrations (Section 4.2) raises concerns about the propagation of model-specific biases or errors. If a model generates flawed reasoning chains, using them as demonstrations could reinforce these flaws in a feedback loop.

Additionally, the use of **self-generated CoT** (Section 4.2.1) where models are prompted with their own potentially incorrect reasoning steps (the “Wrong” set) warrants a safety discussion. While the paper shows this can sometimes improve performance due to distributional alignment, it also risks reinforcing systematic errors or “hallucinated” logic patterns. The authors should briefly address the ethical implications of training models on their own flawed outputs.

Finally, while the paper uses standard benchmarks, the authors should explicitly state that their findings on “reasoning” are limited to the specific tasks evaluated (math, logic puzzles) and may not generalize to real-world reasoning tasks involving nuanced ethical judgment or complex social dynamics, where the risks of automated reasoning are highest.

The paper does not appear to involve human subjects, animal testing, or private data, so no IRB/IACUC concerns are raised. The primary safety gaps are the lack of a robust Impact Statement and the absence of a discussion on the dual-use potential of enhanced reasoning capabilities.

Action items.

- [**writing**] The Impact Statement (Section 7) is generic and dismissive. Given the paper’s focus on reasoning capabilities and the potential for these models to be used in high-stakes domains (e.g., legal, medical, or educational reasoning), the authors must expand this section to explicitly discuss risks of automated reasoning errors, potential for hallucination in critical contexts, and dual-use concerns regarding the optimization of reasoning for malicious tasks.
- [**writing**] The study uses self-generated CoT demonstrations from weaker models (Section 4.2.1) which may contain incorrect reasoning steps. The authors should add a discussion

on the safety implications of training models on potentially flawed or misleading reasoning chains, and whether this could reinforce systematic errors or biases in the model's reasoning process.

- [**writing**] The paper evaluates on datasets like MATH and GSM8K but does not address the potential for these reasoning capabilities to be exploited for generating sophisticated disinformation, social engineering scripts, or automated code generation for malicious purposes. A brief risk assessment regarding the dual-use nature of improved reasoning capabilities is required.

Scientific Evidence

Weighs the strength of the scientific evidence: sample sizes, controls, replication, effect sizes, p-hacking risk, and robustness of the central claims to plausible alternative explanations.

Reviewer model: `qwen.qwen3.5-122b`

Prompt: `agents/prompts/paper_reviewer_scientific_evidence.md`

The paper presents a compelling empirical investigation into the scaling behavior of Many-Shot CoT-ICL, challenging the assumption that more demonstrations always yield stable improvements. The distinction drawn between reasoning and non-reasoning tasks, and between reasoning-oriented and standard LLMs, is well-supported by the scaling curves in Figures 1 and 2. The evidence that similarity-based retrieval fails on reasoning tasks (Section 4.2) is particularly strong, supported by both aggregate trends and the qualitative example in Appendix Table 1.

However, the statistical robustness of the central claims regarding the proposed CDS method and the curvature correlation requires strengthening. First, the reported performance gains of CDS (e.g., the 5.42% gain mentioned in the abstract and Table 1) are presented as point estimates. Given the paper's own finding that performance variance increases with the number of demonstrations (Section 4.4), the absence of statistical significance tests (e.g., paired t-tests or bootstrap confidence intervals) for the CDS vs. Origin comparisons is a critical gap. Without this, it is difficult to distinguish genuine methodological improvement from random fluctuation in the prompt ordering.

Second, the correlation analysis in Section 4.3.2, which links embedding-space curvature to accuracy, is based on only five random permutations per task. While the reported correlation coefficients (e.g., r=-0.628) appear strong, a sample size of N=5 is statistically fragile and highly susceptible to outliers. A more robust validation would involve sampling a larger number of random orderings (e.g., 20-50) to establish the stability of this correlation.

Finally, the "procedural-corruption" ablation in Table 2, while conceptually sound, uses a static CoT from the first demonstration for all subsequent examples. This design introduces a confound: the model is not just receiving "mismatched" procedures, but also a highly repetitive, non-diverse context. It is unclear if the performance drop is due to the procedural mismatch or the lack of demonstration diversity. A control condition using shuffled original CoTs or CoTs from a semantically distinct but structurally similar task would better isolate the specific contribution of procedural alignment.

Action items.

- **[science]** The claim that CDS yields 'up to a 5.42 percentage-point gain' (Abstract) lacks statistical significance testing. Table 1 shows point estimates, but without p-values or confidence intervals for the differences between 'origin' and 'CDS', it is unclear if these gains exceed random variance, especially given the observed order-scaling variance in Section 4.4.
- **[science]** The correlation analysis in Section 4.3.2 (r=-0.547) relies on only 5 random permutations per task to establish the curvature-performance relationship. This sample size is insufficient to robustly claim a strong negative correlation or to rule out spurious associations driven by the specific dataset splits used.
- **[science]** The 'procedural-corruption' ablation (Table 2) uses a static CoT from the first demonstration for all examples. This conflates 'procedural mismatch' with 'repetition bias' or 'loss of diversity'. A stronger control would involve shuffling the original CoTs or using CoTs from a different task to isolate the effect of procedural alignment specifically.

Statistical Analysis

Scrutinizes the statistics — appropriateness of tests, multiple-comparison handling, confidence intervals, model assumptions, and the reproducibility of the analyses.

Reviewer model: `qwen.qwen3.5-122b`

Prompt: `agents/prompts/paper_reviewer_statistical_analysis.md`

The statistical analysis in this paper is generally sound in its experimental design, particularly the use of multiple random seeds (n=5) to assess order sensitivity and the inclusion of standard deviation in the results tables (Appendix B, Tables 1-3). However, the manuscript relies heavily on visual trends and point estimates without providing formal statistical significance testing for its key claims.

First, in Section 4.2, the conclusion that "self-generated demonstrations consistently outperform dataset-provided CoTs" is drawn from Figures 3 and 4. While the trends are visible, the text does not report p-values or confidence intervals for these differences. Given the small sample size (5 seeds), it is crucial to verify that the observed performance gaps (e.g., the ~3-5% gains) are statistically significant and not within the margin of error. The authors should perform paired t-tests or bootstrap resampling on the seed-level results to support these claims.

Second, Section 4.3 introduces a correlation analysis between embedding curvature and accuracy, reporting specific Pearson coefficients (e.g., r = -0.547). While the direction of the correlation is informative, the manuscript omits the associated p-values and confidence intervals. Without these, it is impossible to determine if the correlation is statistically distinguishable from zero, especially given the limited number of random permutations used to generate the data points.

Finally, the ablation study in Section 5.1 (Table 4) compares CDS against a high-curvature baseline. The text asserts that CDS "consistently outperforms" this baseline. To substantiate this, the authors should report the results of a statistical test (e.g., a paired t-test across the different tasks and models) to confirm that the improvements are significant. The current presentation of mean values alone is insufficient to rule out random variation, particularly for the smaller gains observed in some settings. Adding these statistical validations would significantly strengthen the empirical claims.

Action items.

- [**science**] In Section 4.2 (Principle 1), the claim that self-generated CoTs outperform ground-truth CoTs relies on visual inspection of Figures 3 and 4. The text must explicitly report the statistical significance (e.g., p-values from paired t-tests or bootstrap confidence intervals) of these differences to rule out random variance, especially given the small number of seeds (n=5) mentioned in Appendix B.
- [**science**] Section 4.3 reports Pearson correlation coefficients (e.g., r=-0.547) between curvature and accuracy. The manuscript must provide the corresponding p-values and confidence intervals for these correlations to establish that the observed relationships are statistically significant and not artifacts of the specific random permutations sampled.
- [**science**] Table 3 and Table 4 report mean and standard deviation across five random seeds. For the CDS method, which is deterministic given a set, the variance should ideally be reported across multiple random *initializations* of the TSP heuristic or multiple random *subsets* of demonstrations, rather than just the ordering seeds used for the baseline. Clarify the source of variance for the CDS results.
- [**science**] The 'High-curvature control' experiment in Section 5.1 compares CDS against a high-curvature baseline. The text claims CDS 'consistently outperforms' this baseline. A formal statistical test (e.g., paired t-test across tasks/models) should be reported to confirm that the observed gains (e.g., the 5.42 pp gain mentioned in the abstract) are statistically significant and not due to chance.

Writing Quality

Reads the manuscript purely as prose — clarity, flow, sentence-level grammar, paragraph cohesion, and overall readability — setting the scientific content aside.

Reviewer model: `qwen.qwen3.5-122b`

Prompt: `agents/prompts/paper_reviewer_writing_quality.md`

The manuscript presents a compelling study on many-shot Chain-of-Thought In-Context Learning (CoT-ICL), but the writing quality requires minor revisions to ensure grammatical precision and readability. While the overall flow is logical and the argument structure is clear, there are several recurring grammatical errors, specifically subject-verb agreement issues and typos, that detract from the professional polish of the paper.

In the Introduction, the authors state, "our experiment uncover," which is a subject-verb agreement error; it should be "our experiments uncover" or "our experiment uncovers." Similarly, in the Related Works section, the sentence "Auto-CoT... propose an automatic few-shot CoT prompting method" incorrectly uses the plural verb "propose" for the singular subject "Auto-

CoT"; it should be "proposes." These errors, while not obscuring the meaning, suggest a lack of final proofreading.

The Conclusion section contains a more significant syntactic issue: "we introduce CDS, a rapid, low cost customization demonstrations orders by minimizing conceptual curvature." This phrase is grammatically incoherent. It likely intends to read "a rapid, low-cost method for customizing demonstration orders" or "a rapid, low-cost customization of demonstration orders." The current phrasing makes the sentence difficult to parse.

Additionally, there are minor typographical errors scattered throughout the appendices. For instance, in the "CDS: Details and Implementation" section, the text ends with "provided CoT..", featuring a double period. In the caption for the figure in Appendix A (selfgen_non_reasonng.pdf), the filename typo "non_reasonng" is repeated in the caption text, which should be corrected to "non_reasoning."

Finally, the abstract contains a slight awkwardness in the phrase "suggests two principles," where the subject "We interpret these behaviors... and suggests" creates a parallelism error. It should be "and suggest" to match the plural subject "We."

Addressing these specific grammatical and typographical issues will significantly improve the manuscript's readability and professional presentation without altering the scientific content.

Action items.

- [**writing**] In the Introduction, the sentence 'our experiment uncover' contains a subject-verb agreement error. It should be 'our experiments uncover' or 'our experiment uncovers'.
- [**writing**] In the Related Works section, the sentence 'Auto-CoT... propose an automatic few-shot CoT prompting method' has a subject-verb agreement error. 'Auto-CoT' is singular, so it should be 'proposes'.
- [**writing**] In the Conclusion, the phrase 'rapid, low cost customization demonstrations orders' is grammatically incoherent. It likely intends to say 'rapid, low-cost customization of demonstration orders'.
- [**writing**] In Appendix section 'CDS: Details and Implementation', the phrase 'under this setup, we study models that can interpret and leverage the provided CoT..' contains a double period typo.
- [**writing**] In the Appendix, the caption for Figure 1 (selfgen_non_reasonng.pdf) contains a typo: 'non_reasonng' should be 'non_reasoning'.